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The effects of conventional and unconventional monetary policy shocks on the stock market

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Abstract
This paper applies weekly data to investigate the effects of the conventional and unconventional monetary policy shocks on US stock returns, using a multivariate structural VAR model that considers transmission mechanisms in channeling the effects of these shocks into the stock market. We identify both types of policy shocks using high and low volatility in the relevant policy variables. Our empirical results suggest that the interest rate channel, not the bank lending channel, plays a major role in the process of transmission and the US stock market responds less to the conventional monetary policy than the unconventional one. We estimate that expansionary policy shocks that lower the long term interest rate by 25 basis points increase stock returns by 0.57% and 2% under the conventional and unconventional monetary policies, respectively, over five weeks.

Keywords Unconventional and conventional monetary policy · Structural VAR model · Heteroscedasticity

JEL Classification E44 · E52 · E58

1 Introduction

Monetary policy has always been considered as the most important factor affecting stock returns. Before 2008, the Fed primarily depends on its conventional policy tool, the federal funds rate, to reach its twin objectives of controlling inflation and maintaining stable employment. However, in the aftermath of the financial crisis in 2008, when the Federal Funds Rate reaches zero lower bound, the Fed started to heavily rely on its unconventional policy tool, changes in the size of its balance sheet, to steer the economy on its right path. The primary objective of this paper is to make

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a comparison between the effects of two types of monetary policy shocks on the US stock market.

Several papers have already investigated the responses of financial markets in the US, Euro, and other areas to shocks to monetary policies. Based on surprises in conventional and unconventional monetary policies, Haitsma et al. (2016) use event study approach to investigate the impact of the ECB's monetary policies on stock markets and find that unconventional monetary policy surprises have stronger effects on EURO STOXX 50 index than conventional policy surprises. In another paper, Claus et al. (2018) also find stronger effects of unconventional policy shocks on a range of US financial assets using a latent factor model that is identified with policy announcement and non-announcement dates. On the other hand, Weale and Wieladek (2022) do not find any statistically significant differences in the financial effects of the quantitative easing (QE) and conventional policy in the Euro area, UK, and US, based on a Bayesian VAR. In comparing the size of the effects, as a benchmark, they use the amount of inflation generated by the policies.

In this paper, we reexamine the effects of conventional and unconventional monetary policy shocks on US stock returns, using a multivariate structural model that is different from those in earlier papers, as it builds on two types of possible channels–bank lending and interest rate channels–in transmitting the policy shocks into the stock market. We take into account these transmission channels in setting up the model, as both types of policies appear to be effective in affecting stock returns by changing the availability of commercial loans and rebalancing the investors' portfolio. The model is estimated for each of the policy regimes and in each case policy shocks are identified using volatility in the policy tools, following empirical approaches as detailed in Rigobon (2003) and Wright (2012). We are specifically interested about their modeling strategies over others, as they are able to capture the dynamics of the high frequency daily/weekly data in a vector regressive framework, using high and low volatility of the policy variables to identify the model. We use federal funds rate and changes in the size of the Fed's balance sheet as conventional and unconventional policy tools, respectively. We use the estimated model to calculate impulse responses of stock returns to shocks to monetary policies and, in doing that, we evaluate both channels in transmitting the shocks. Based on this evaluation, we finally compare between the effects of conventional and unconventional monetary policy shocks on US stock returns.

In addition to the modeling framework, our paper is also different from others in terms of the frequency of the dataset. In choosing this frequency, we find that the weekly dataset is the most suitable one to serve our purpose, as it is the highest level (of frequency) that allows us to incorporate transmission mechanism as mentioned before. Moreover, compared to the high frequency daily data, our lower frequency weekly data works much better to capture persistent effects, as explained in Gurkaynak et al. (2022). We understand that there is an issue of endogeneity associated with using lower frequency data, as there might be a feedback response of monetary policy to the changes in stock returns. Our dynamic structural VAR, however, is able to address this issue. In the context of using weekly dataset, our paper is also different from Rahman and Serletis (2023) that uses the same frequency, but applies a simple bivariate structural model, to investigate the relationship between unconventional

monetary policy and US stock returns, without paying any attention to the transmission of the shocks and comparison between the effects of two types of policy shocks (conventional and unconventional) on stock returns. The transmission mechanism that we introduce in this paper poses a unique challenge in estimating the model over two policy regimes and, then, interpreting empirical results. In sum, we consider this paper as a major improvement over Rahman and Serletis (2023) in terms of research questions, specification of the empirical model, and findings.

Our empirical results show positive effects on stock returns due to unanticipated expansionary shocks to both types of monetary policies, suggesting that these policies, in general, have statistically significant effects on the US stock market. In analyzing the transmission of these shocks, our findings offer consistent evidence in favor of the interest rate channel over the bank lending channel, regardless of the policy regimes. In addition, we find stronger effects of the unconventional monetary policy on the US stock returns than the conventional one. In fact, we document 0.57% and 2% increases in stock returns under the conventional and unconventional monetary policy environments, respectively, over five weeks, due to expansionary policy shocks that lower the long term interest rate by 25 basis points. Although not reported in the paper to conserve space, we find that our empirical model is well specified, as it shows reasonable responses of monetary policies to unanticipated changes in stock returns. Finally, we extend our analysis at somewhat disaggregate levels, by estimating the responses of portfolio returns to conventional and unconventional monetary policy shocks. This supports our main empirical results.

In specifying our empirical model, we understand that a number of papers, including Gertler and Karadi (2015), Gorodnichenko and Weber (2016), and Swanson and Jayawickrema (2024), argue that high frequency changes in interest rates around policy announcements should be used to identify the monetary policy shocks. In this regard, we point out that the frequency of our dataset is weekly instead of daily, and we use volatility in the policy tools to identify the monetary policy shocks based on the evidence that the volatility changes during the sample periods. It is also important to note that our research focuses on ex post, not ex ante, effects of monetary policies on the stock market. It is well established in the literature that although the stock market responds right after the policy announcements, it continues to do so during and after implementation of the policies, because of unforeseeable changes in economic fundamentals that happen after the announcements. Mamaysky (2018), for example, shows the long horizon responses of stock prices due to quantitative easing (QE) announcements by the FED, ECB, and the Bank of England. In addition, Kuttner (2001) explains that markets may adjust overtime to the unexpected component of the announced policy that may not be priced into the market right after the announcement. The ex post effects of monetary policies are reflected in the specification of our empirical method that heavily relies on the transmission mechanism on transmitting the effects of the policy shocks that are well identified as they generate reasonable response to the variables of interest.

The outline of the paper is as follows. In Section 2 we discuss the econometric methodology and in Section 3 we present the empirical model and explain the dataset used to estimate the model. In Section 4 we present the empirical results while the final section briefly concludes.

2 Econometric methodology

We use the following reduced form VAR model in order to investigate the dynamic effects of the monetary policy shock on stock returns,

$$z_t = \boldsymbol{C}_s + \sum_{j=1}^{p} \Phi(j) z_{t-j} + \boldsymbol{u}_t, \tag{1}$$

where z_t is an $n \times 1$ vector of variables, $\boldsymbol{C}_s$ is an $n \times 1$ parameter vector, $\Phi(j)$, $j = 1, \ldots, p$, is an $n \times n$ parameter matrix, and $\boldsymbol{u}_t \sim (\mathbf{0}, \Omega)$, with Ω being the variance-covariance matrix.

We assume that the reduced form errors, $\boldsymbol{u}_t$, are related to an $n \times 1$ vector of underlying independent structural shocks, $\boldsymbol{e}_t$, as follows

$$\boldsymbol{u}_t = \sum_{i=1}^{n} \boldsymbol{B}_i \boldsymbol{e}_t, \tag{2}$$

where $\boldsymbol{B}_i = \left[b_{1i} \; b_{2i} \cdots b_{ni} \right]'$.

In order to identify a shock to any of the variables, we follow Wright (2012) and consider two periods of variances in that particular variable, high and low. We assume that the shock has mean zero and variance σ_l^2 in low variance periods and σ_h^2 in high variance periods. We also assume that $\sigma_l^2 \neq \sigma_h^2$ and that all other structural shocks are identically distributed with mean zero and variance one in all periods. If Ω_l and Ω_h are used to denote the variance-covariance matrix of the reduced form errors in low and high variance periods, respectively, then (since we are interested in identifying the shock to the variable ordered first)

$$\Omega_h - \Omega_l = \boldsymbol{B}_1 \boldsymbol{B}_1' (\sigma_h^2 - \sigma_l^2) \tag{3}$$

where Eq. 3 identifies B_1 if the normalization $\sigma_h^2 - \sigma_l^2 = 1$ is imposed.

In order to estimate the parameters in $\boldsymbol{B}_1$, we estimate Eq. 1, construct the sample variance-covariance matrices, $\hat{\Omega}_h$ and $\hat{\Omega}_l$, and solve the following minimum distance problem

$$\hat{\boldsymbol{B}} = \arg\min_{\hat{\boldsymbol{B}}_1} [\text{vech}(\hat{\Omega}_h - \hat{\Omega}_l) - \text{vech}(\boldsymbol{B}_1 \boldsymbol{B}_1')]' [\hat{\boldsymbol{V}}_l + \hat{\boldsymbol{V}}_h]^{-1} [\text{vech}(\hat{\Omega}_h - \hat{\Omega}_l) - \text{vech}(\boldsymbol{B}_1 \boldsymbol{B}_1')], \tag{4}$$

where $\hat{\boldsymbol{V}}_h$ and $\hat{\boldsymbol{V}}_l$ are estimates of the variance-covariance matrices of $\text{vech}(\hat{\Omega}_h)$ and $\text{vech}(\hat{\Omega}_l)$, respectively. In solving the model, we use an autoregressive lag length of thirty weeks based on the Akaike Information Criterion.

The impulse response functions that show the effects of the monetary policy shock on stock returns can be calculated in the usual way. We use the bootstrap procedure described in Wright (2012) to calculate the confidence interval.

3 Data and the macroeconomic model

Based on our econometric specification, we set up the following reduced form macroeconomic VAR that includes an indicator of monetary policy (mp_t), commercial and industrial loans (cr_t), the growth in dividend yields (dv_t), an indicator of the long-term interest rate (lr_t), and stock returns (ret_t),

$$\begin{bmatrix} mp_t \\ cr_t \\ dv_t \\ lr_t \\ ret_t \end{bmatrix} = \boldsymbol{C}_s + \sum_{j=1}^{p} \Phi(j) \boldsymbol{z}_{t-j} + \boldsymbol{u}_t \tag{5}$$

This macroeconomic model is separately estimated over two policy regimes, 1973-2007, and 2008-2021, as the Fed used two different types of policy instruments during those periods. Our empirical approach in this regard follows existing literature that identifies a structural change in the Federal Reserve's monetary policy around 2008. Swanson (2021), for example, applied a factor model to analyze the conventional and unconventional policy components of the FOMC announcements and found that, before 2008, the Fed mostly relied on its conventional monetary policy tool, the federal funds rate, to steer the economy. On the other hand, the Fed largely followed an unconventional policy approach after 2008, by changing the size of its balance sheet, as the federal funds rate reached the zero lower bound. In another paper, Inoue and Rossi (2021) analyzes daily zero coupon bond yields at different maturities and reported similar dating of policy changes around 2008, as monetary policy primarily works through interest rates. In the context of applying our econometric framework to two different sample periods, these two papers serve as precedence, as, after identifying the policy regimes, they measure their empirical models for each of the regimes to investigate the effects of the Fed's conventional and unconventional monetary policies on economic and financial variables. Thus, following this recent literature on identifying, measuring, and comparing the Fed's policy effects, we consider our approach very reasonable. It is further to be noted here that, in order to compare the magnitude of responses of relevant variables over two types of policy shocks, we allow long-term interest-based normalization, as described bellow.

Our rationale for the choice of sample periods is also very much evident in Fig. 1 that shows weekly data on the effective Federal Funds Rate for 1973-2021 and logged total assets for 1990-2021 (on the secondary axis). Note that the series on the Fed's total assets is only available from 1990. As Fig. 1 shows, the federal funds rate ceased to work as an effective monetary policy tool at the end of 2007 when it reached the zero lower bound and remained there for a long period. Around the same time, we observe a large spike in total assets, although the Fed mostly maintained it at a steady level before 2008. In the event of the bottom low interest rate, the balance sheet remains to be an alternative tool for the Fed during 2008-2021 to move the economy further towards its dual mandates. There was, however, a brief interruption in the unconventional policy during late 2016 to the beginning of 2020, when short-term interest rates became positive. We, therefore, re-estimated the model for the

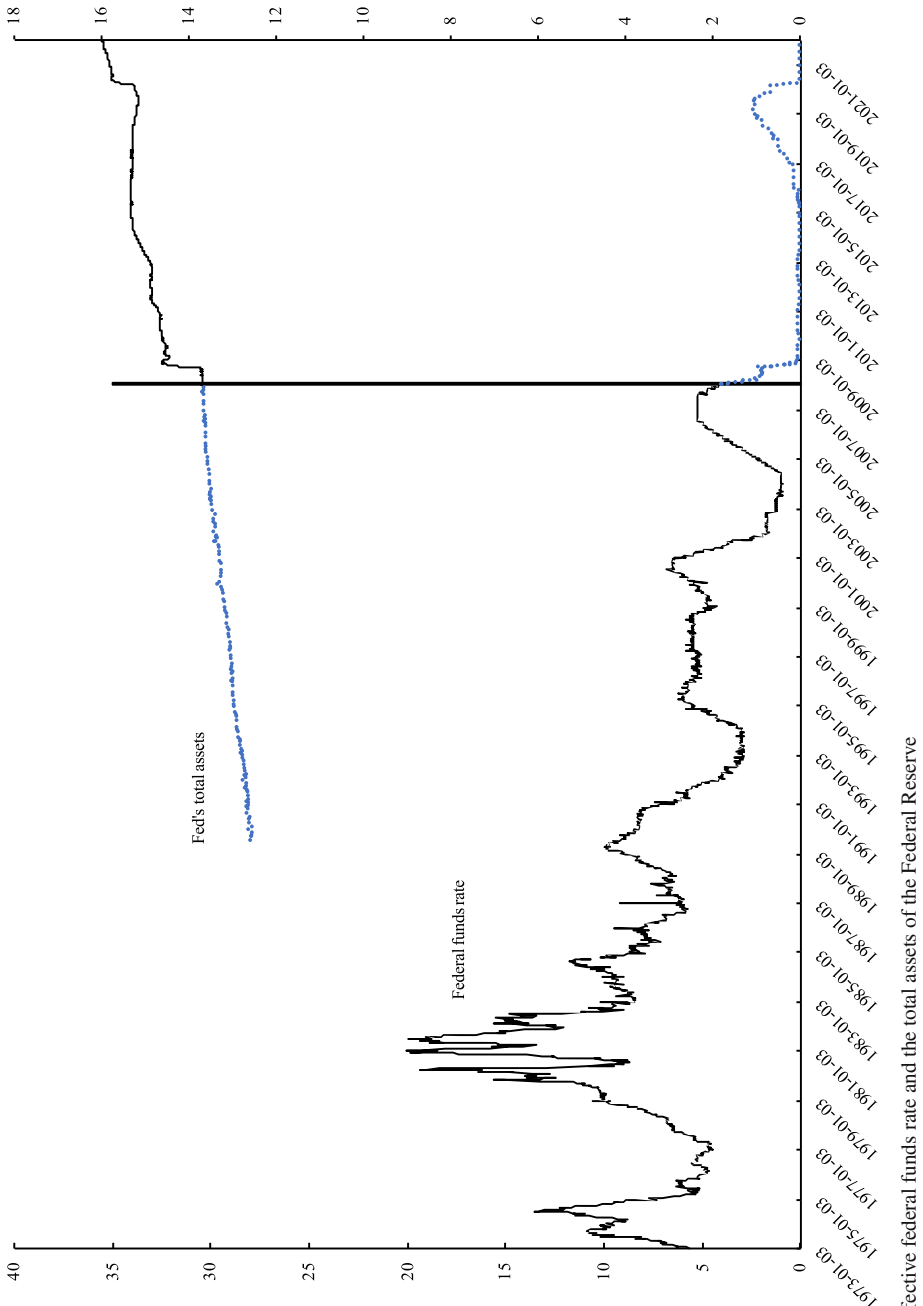


Fig. 1 Effective federal funds rate and the total assets of the Federal Reserve

unconventional policy after dropping those sample periods, but could not find any major differences in our empirical results.

The data on the Fed's balance sheet are collected from the FRED database. We use data on U.S. market returns available on the Center for Research in Security Prices (CRSP). This return data is based on stock prices of all firms listed on the NYSE, AMEX, or NASDAQ and considered as the most comprehensive. We follow Torous et al. (2004) to construct the real dividend yields from the weekly returns on the value-weighted market portfolio with and without dividend that are available on CRSP database. We use the yields on 10-year treasury securities as an indicator of the long-term interest rate. The data on commercial and industrial loans, yields, and the federal funds rate are all available on the FRED database. Finally, the portfolio returns series, that we use to extend our analysis, are collected from Kenneth French's online data library.[1] It is to be noted here that our main empirical results remain unchanged if we change the ordering of the variables in Eq. 5.

In setting up our empirical model, we consider possible transmission mechanisms on the effects of the monetary policy shocks into the stock market. Both types of policy shocks have considerable impact on the reserves of commercial banks and, thus, affect their ability to generate all types of loans including commercial and industrial loans. This is known in the literature as the bank lending channel. Bernanke and Gertler (1995) first identified this channel in explaining the transmission of monetary policy shocks in the economy, as it affects the external finance premium—a difference in costs for firms between external and internal finances. A large literature exists on identifying and testing the bank lending channel—see, for example, Gambacorta et al. (2011) and Jiménez et al. (2020). In particular, in our empirical framework, an expansionary conventional policy shock that reduces short-term interest rates involves the Fed in purchasing assets through open market operations. This leads to an increase in the amount of available reserves for commercial banks. Similarly, the Fed's large scale asset purchases during unconventional policy periods supply a large amount of reserves to the banking system. Thus, in the course of correcting the economy by generating more or less loans at a higher or lower interest rate, the Fed's monetary policy in effect creates variable opportunity for firms to earn profits that are distributed as dividends to the stockholders. Our modeling framework, therefore, takes into account the bank lending channel by including dividend yield, in addition to the total amount of commercial and industrial loans.

Besides the bank lending channel, we consider long-term interest rates as a possible medium to carry the effects of both types of policy shocks into the stock market. It is well known to the market participants that one of the objectives of the unconventional monetary policy is to reduce long-term interest rates using the "portfolio balance" effect, as the short-term rate is already very low. This is based on the assumption that securities in investors' portfolios are not perfect substitutes due to preferred habitat/market segmentation and, therefore, the large scale asset purchases reduce yields by altering duration risk in the market. For a review on the literature of the portfolio balance effect—see Gagnon et al. (2011), Joyce et al. (2011), and Neely (2015), among others. On the other hand, the theories of the term-structure of interest rates suggest a

[1] http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/index.html

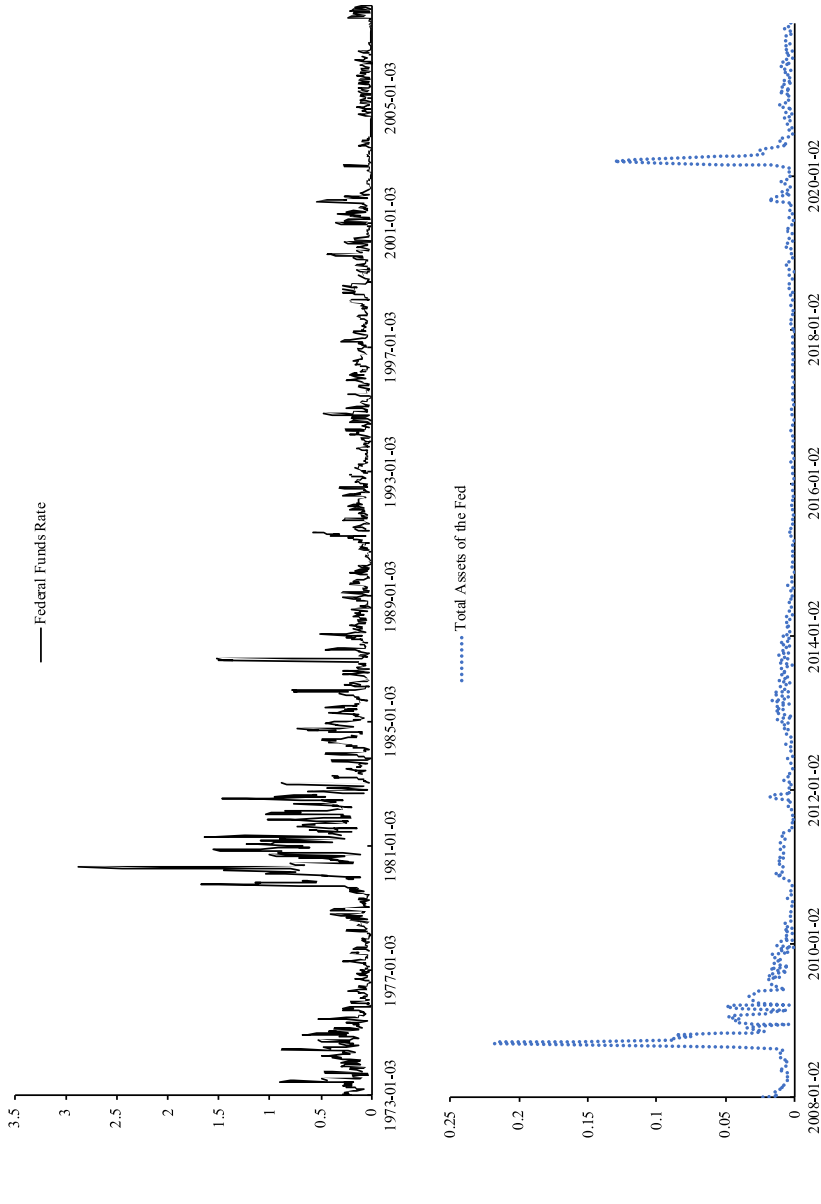


Fig. 2 Volatility of the federal funds rate and total assets of the Federal Reserve

co-movement between short and long-term rates during conventional policy periods, when short-term rates remain at a non-zero level. All of these indicate that investors usually experience changes in the current yields of the assets with long maturity under both types of monetary policy, which induces them to adjust their portfolios and, thus affects stock returns, by reallocating investments between the bond and stock markets. We capture the interaction of the policy variables and long-term interest rates and how it affects stock returns, by including 10-year bond yield in Eq. 5 as a suitable indicator of the long-term interest rate.

Finally, following our econometric framework in Section 2, we identify conventional and unconventional monetary policy shocks using volatility in the federal funds rate and the size of the Fed's balance sheet, respectively. In order to show the level of fluctuations for the policy variables and how it helps on identification, we plot 4-weeks of rolling standard deviations in Fig. 2 for each of the series on the effective federal funds rate as well as total assets of the Federal Reserve, for 1973-2007 and 2008-2021, respectively. Although we find a considerable level of volatility in both series, its magnitude changes from period to period. The federal funds rate, for example, shows high level of volatility in the early part of the sample period, especially in the Volcker era, and gradually diminish at the end. Fed's balance sheet, on the other hand, shows substantial level of fluctuations in the beginning and ending of the sample periods, when the Federal Reserve was aggressively engaged in purchasing different types of assets. Since we are using weekly data, following Rigobon and Sack (2003), periods of high/low volatility are defined as when the four-weeks rolling variance of the residuals of the policy indicator is more/less than one standard deviation above its average.

4 Empirical evidence

In this section, we first discuss the effects of each type of monetary policy shocks on stock returns. Our ultimate objective is to compare the effects of the conventional monetary policy shocks on stock returns with those of the unconventional policy shocks, based on the transmission mechanism that we explained before. Finally, we use our empirical model to analyze responses of portfolio returns to policy shocks in order to check further evidence for our main findings.

4.1 Effects of monetary policy shocks on stock returns

We apply our identification strategy to estimate Eq. 5 for the conventional monetary policy period, 1973-2007, using the federal funds rate as a policy variable. We then calculate responses of the variables to a one standard deviation negative shock to the policy rate. The results are displayed in Fig. 3 with a 84% confidence interval. Panel (D) shows a statistically significant positive effect of the expansionary policy shock on stock returns. In particular, in response to a one standard deviation shock to the policy rate, stock returns increase by 0.05% on impact and show a 0.10% cumulative increase over a five-week period.

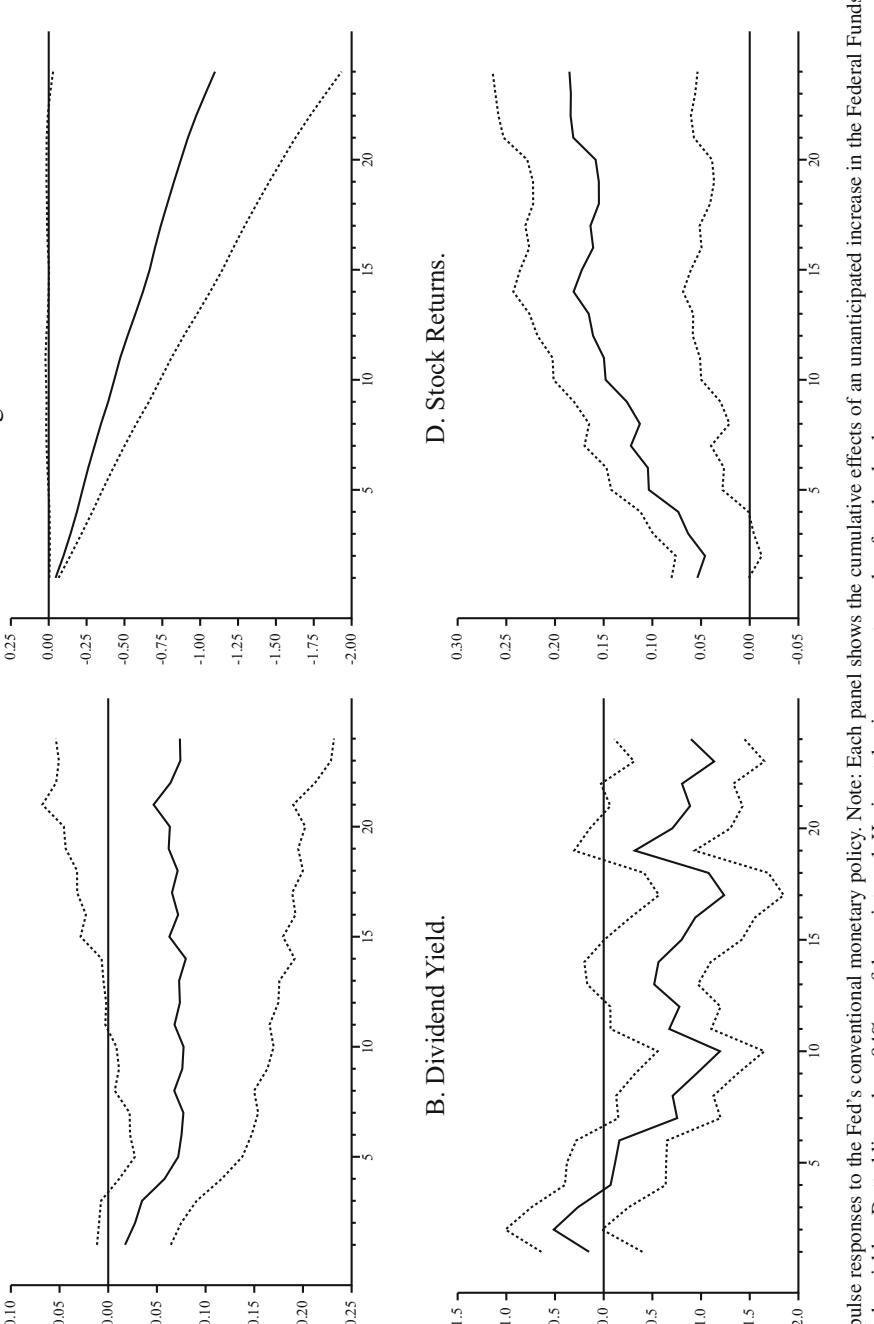


Fig. 3 Impulse responses to the Fed's conventional monetary policy. Note: Each panel shows the cumulative effects of an unanticipated increase in the Federal Funds Rate on indicated variables. Dotted lines show 84% confidence interval. Horizontal axis represents weeks after the shock

Regarding responses of the other variables, we find that an unanticipated drop in the federal funds rate significantly decreases the long-term interest rate. As shown in Panel (C), a one standard deviation conventional negative policy shock in our model decreases the 10-year bond yield by 0.18% in a month. In this regard, it is to be noted here that the responses of the long-term interest rate are not very persistent throughout the horizon, as the confidence interval sometimes includes zero. On the other hand, total amount of commercial loans in Panel (A) displays an insignificant response to a conventional policy shock at the beginning of the horizon and, then, a cumulative decline for almost seven weeks. Finally, after an expansionary policy shock, Panel (B) shows a brief positive response in the growth of dividend yields in the second week of the horizon before it turns out to be statistically insignificant for the next four weeks.

The drop in the long-term interest rate due to a negative shock to the short-term rate is expected, as it follows the theory of the term structure of interest rates. The response of commercial loans, however, does not fall into the line with what has usually been reported in the literature. This might be explained as follows. In an attempt to lower the short-term interest rate, although the Fed's policy of open market purchases supplies more reserves to the overnight market and creates an opportunity for banks to generate more loans, it is counterbalanced by the reduction in deposits on the liability side of the banks' balance sheet due to low deposit rates. Moreover, as the monetary policy shock has substantial effects on the demand side of the economy, the commercial banks might reallocate their resources to generate more consumer loans than business loans. The reduced amount of commercial loans limits the firms' capacity to expand their business and, therefore, we do not find any significant effect of the policy shock on the growth of dividend yields.

We next substitute the Federal Funds Rate in Eq. 5 using the size of the Fed's balance sheet as a policy variable and apply our identification strategy to estimate Eq. 5 for the unconventional policy period, 2008-2021. As before, we calculated responses of the variables to a one standard deviation positive shock to the unconventional policy variable and show these in Fig. 4 with a 84% confidence interval. In choosing sign of the shock, we picked the positive one in order to compare the results later with those of the conventional policy.

Similar to the the effects of the conventional policy shock, Fed's unconventional expansionary policy boosts the stock market by having a significant positive effect on stock returns. As shown in Panel (D) of Fig. 4, there is a cumulative increase of about 0.20% in stock returns in the first five weeks after an unexpected increase in the size of the balance sheet by one standard deviation. This expansion in balance sheet results in a cumulative decrease of 0.15% in the long-term interest rate over five-weeks, as can be found in Panel (C). We observe the same low level of persistence in responses of the long-term interest rate to the unconventional policy shock as we reported before for the conventional one. We find persistent positive effects of the unconventional policy shock on the total amount of commercial loans in Panel (A). However, the response of the growth in dividend yields in Panel (B) remains statistically insignificant for the first seven weeks.

Our findings in Fig. 4 suggest that the unconventional monetary policy shock produces empirical results that appear to be quite reasonable. The decrease in the long-term interest rate due to the expansionary policy is expected, as one of the goals of quan-

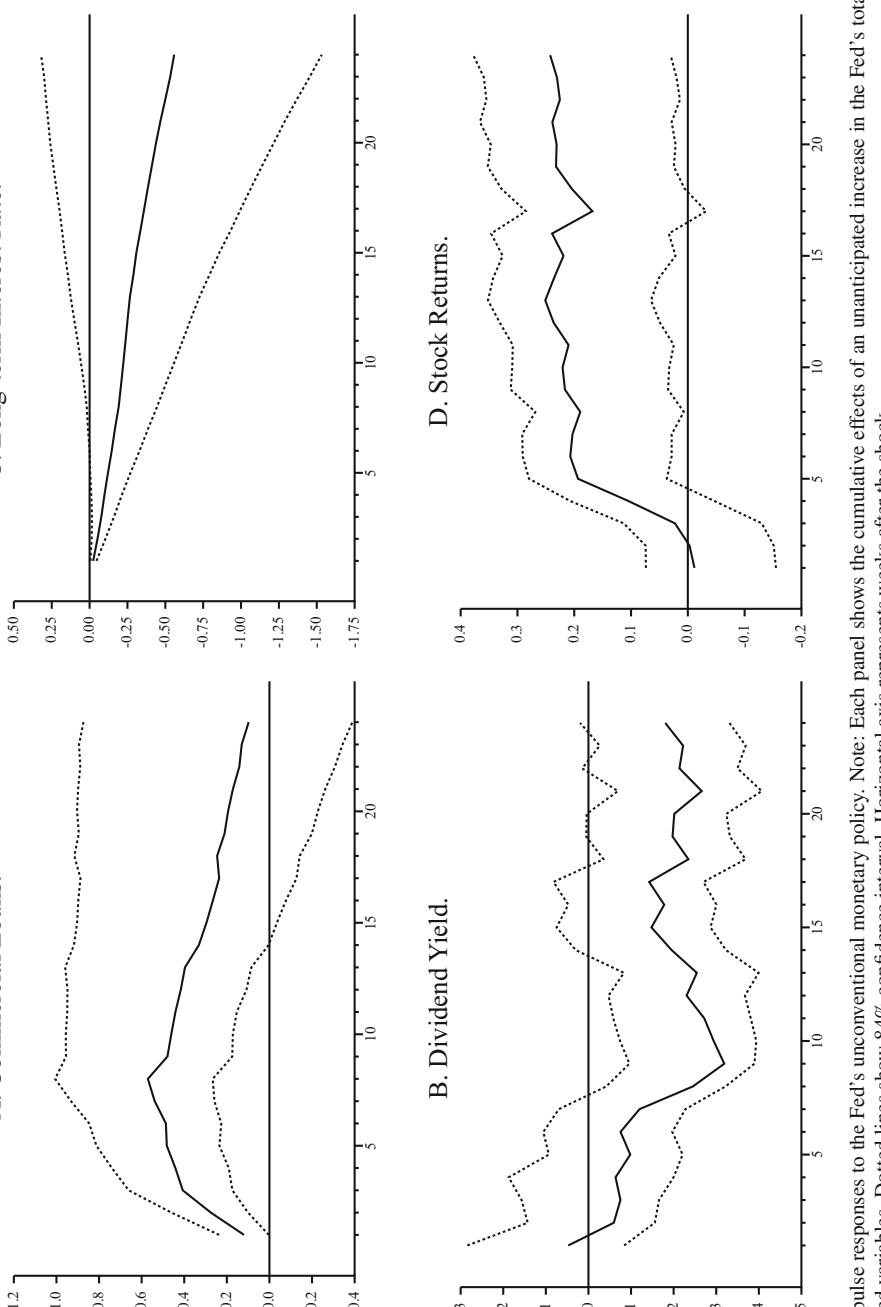


Fig. 4 Impulse responses to the Fed's unconventional monetary policy. Note: Each panel shows the cumulative effects of an unanticipated increase in the Fed's total assets on indicated variables. Dotted lines show 84% confidence interval. Horizontal axis represents weeks after the shock

titative easing, as administered by the Fed, is to lower the long-term interest rate when the short-term rate already reaches the zero lower bound. On the other hand, the systematic increase in the total amount of commercial loans, as opposed to the insignificant response as reported before in case of the conventional monetary policy, matches intended outcome of the Fed's several rounds of aggressive asset purchase programs that provide a high level of liquidity in the banking sector. This outweighs the commercial banks' loss of deposits due to bottom low deposit rates and allows them to raise the supply of commercial loans to the firms. However, we do not find any significant effects of increasing amount of commercial loans on the growth in dividend yields. This might happen, as firms accepting loans have an intention to accumulate cash reserves, instead of being engaged in real economic activity by expanding their business.

4.2 Comparison on the effects of the monetary policy shocks on stock returns

Finally, we compare between the responses of stock returns to the conventional and unconventional monetary policy shocks. To make a fair comparison, we first assess the effectiveness of the transmission channel, based on the empirical results that were reported in the Section 4.1. In general, they do not lend support to the bank lending channel. As discussed before, an expansionary shock under conventional monetary policy fails to boost total amount of commercial and industrial loans in our model. In case of unconventional monetary policy, although an increase in the size of the Fed's balance sheet has an expected effect on commercial and industrial loans, it is not well translated in the response of the growth in dividend yields. On the other hand, monetary policy shocks in both cases push the 10-year bond yield in the right direction to cause a noticeable impact on stock returns. All of these suggest that the interest rate channel primarily helps to carry over monetary policy shocks into the stock market.

Figure 5 shows the comparison between the effects of the conventional and unconventional monetary policy shocks on stock returns. To make this comparison, we set up the size of the expansionary policy shocks in such a way that the long-term interest rate would decrease by 25 basis points. The broken and solid lines refer to the cumulative responses of stock returns to the conventional and unconventional policy shocks, respectively, with dots indicating significance of responses at the 84% confidence interval. We find persistent effects of both types of policy shocks on stock returns, but conventional monetary policy generates an immediate response in the stock market, while there is a delayed response under the unconventional policy. However, throughout the forecasting horizons, we find a much larger effects of the unconventional monetary policy shocks on stock returns than those of the conventional monetary policy. In particular, an unconventional policy shock that lowers the long-term interest rate by 25 basis points increases stock returns by more than 2% over a 5-week period. On the other hand, over the same period, a conventional policy shock causing the same effect on the long-term interest rate increases stock returns by 0.57%

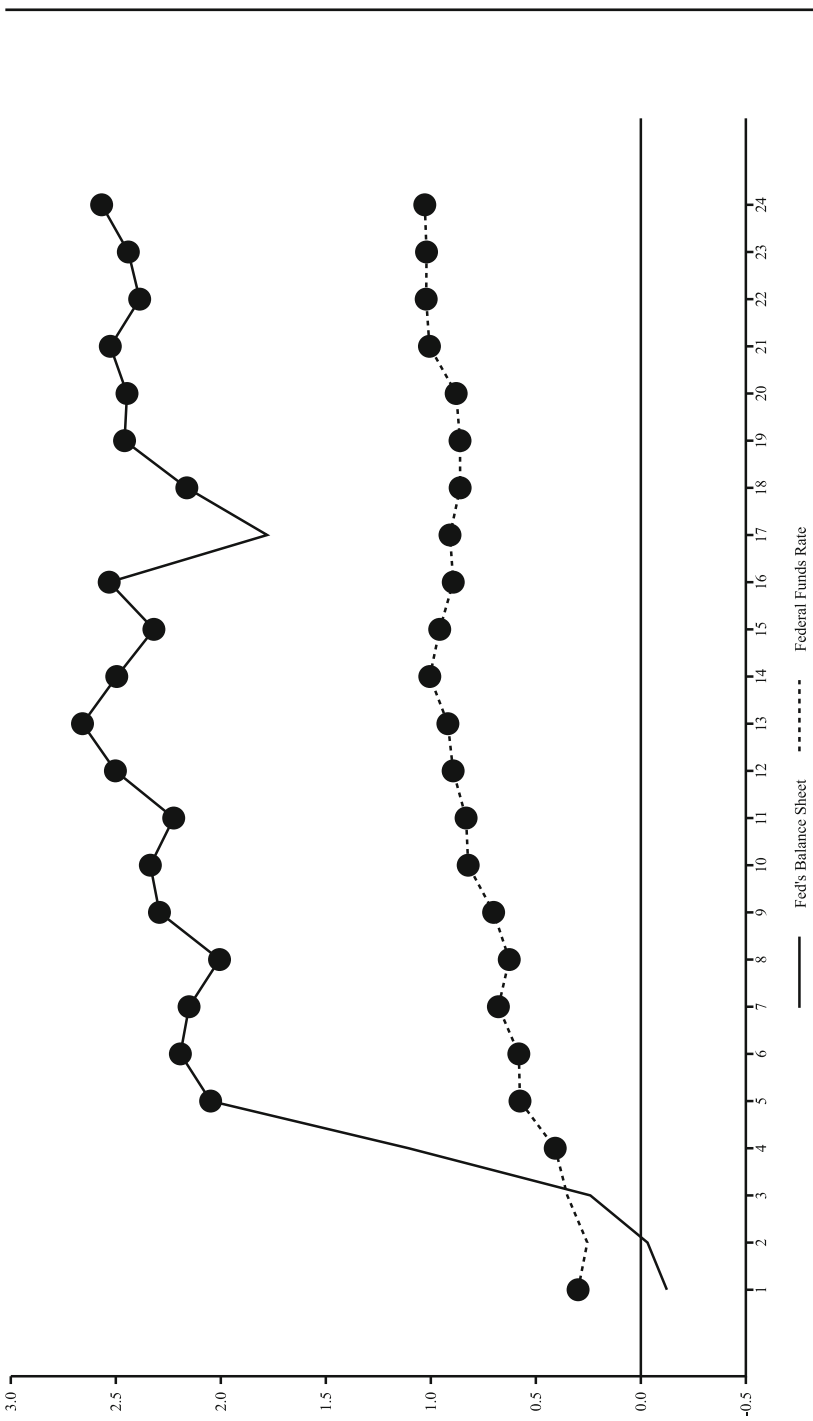


Fig. 5 The effects of conventional and unconventional monetary policy shocks on stock returns. Note: The solid and dotted lines show the responses of stock returns to shocks to the Fed's total assets and the federal funds rate. For comparison, policy shocks are set to decrease the long-term rate by 25 basis points. Dots indicate significance at 84% confidence interval. Horizontal axis represents weeks after the shock

4.3 Portfolio returns

After discussing the effects of two types of monetary policy shocks–conventional and unconventional–on aggregate returns, in this section we extend our analysis to portfolio returns that are based on four different characteristics: firm size, operating profitability, investment, and book to market values. For each of the four characteristics, the portfolios are further segregated into three types: Lo, Med, Large/Hi. We re-estimated the model Eq. 5 in all twelve cases for both policy regimes and calculated the responses due to a one standard deviation shock to conventional and unconventional policies. We report the results under four blocks in Fig. 6. In doing that, in the last row of each block, we compare the results to investigate if a particular policy shock has any differential impact on three types of portfolios of a given characteristic. In the same way, in the last column of the block, we compare the portfolio returns of a specific type over conventional and unconventional policy shocks.

We find that both types of monetary policy have statistically significant effects on portfolio returns in most of the cases, similar to what we already observed before for aggregate returns. In particular, the size, sign, and shape of responses in portfolio returns due to policy shocks are similar to those reported before for aggregate returns. In comparing the responses for the conventional policy shock over Lo, Med, and Large/Hi portfolios, we do not see much differences, as shown in the bottom row of each block. However, in case of unconventional policy, the responses of portfolio returns for firms with small size, low operating profitability, and low book to market values are higher compared to returns for firms with medium and large size as well as returns for firms with medium and high operating profitability and book to market values. This may happen, because firms with smaller size, low profitability, and low book to market values rely on bank based financing instead of market based financing and, as a result, directly benefit more than their medium and large or high counterparts due to quantitative easing, which increases the flow of commercial loans, as we discussed in Section 4.1. On the other hand, regardless of portfolio types (Lo, Med, Large/Hi), the rightmost column in each block clearly indicates larger effects of the unconventional policy shocks than those of the conventional policy shocks. This supports our main findings in Section 4.2.

5 Conclusion

In this paper, we move forward the empirical literature that makes a comparison between the effects of the conventional and unconventional monetary policy shocks on the stock market. Our empirical approach is built around a multivariate structural VAR that includes relevant variables on transmission of the policy shocks into the US stock market through bank lending and interest rate channels. In specifying the empirical model, we primarily focus on implementation, instead of announcements, of the monetary policy. Using a weekly dataset on five variables, the model is identified with volatility of the policy variables and, then, estimated over suitable time-frames for conventional and unconventional monetary policy regimes. We find statistically significant positive effects of both types of expansionary policy shocks on US stock returns.

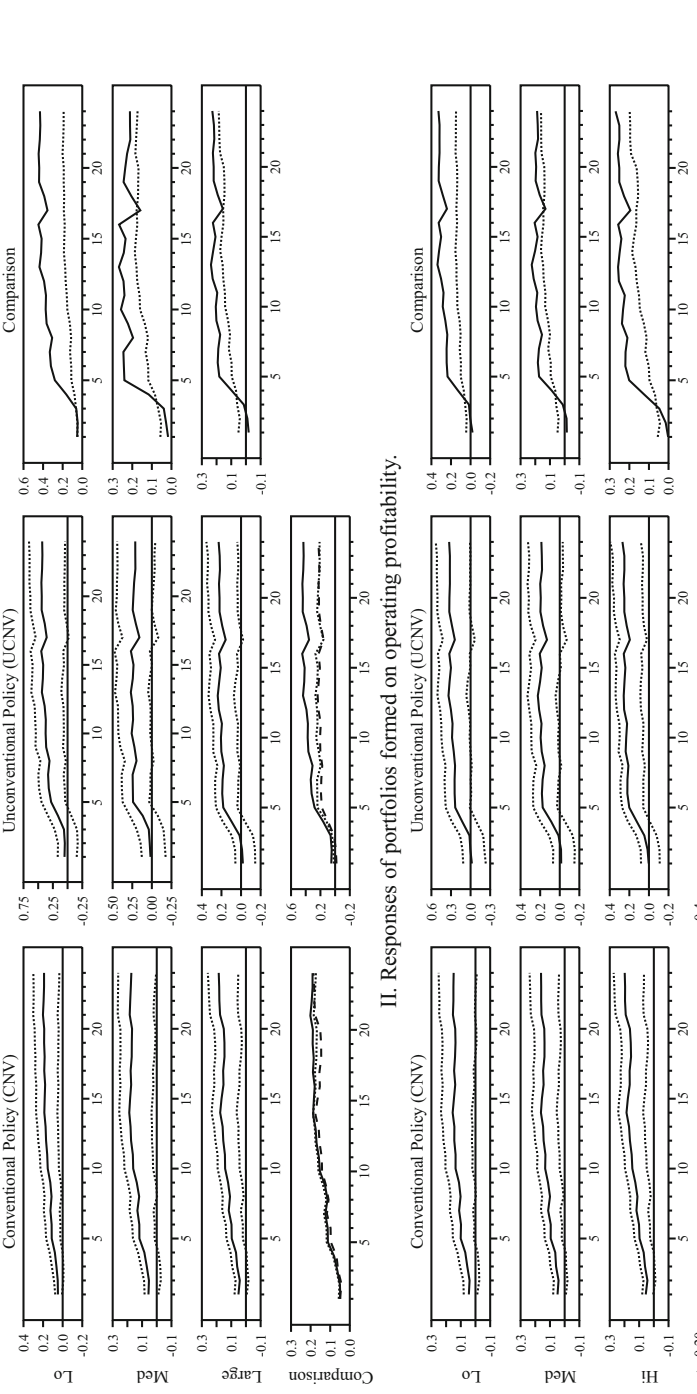


Fig. 6 Comparison on the effects of the conventional and unconventional monetary policy shocks on portfolio returns. Note: Dotted lines in each box except in the last row and column indicate 84% confidence interval. Solid, dotted, and dashed lines in the last row show responses of lo, medium, and large/hi portfolios. Solid and dotted lines in the last column refer to unconventional and conventional policy shocks. Horizontal axis represents weeks after the shock

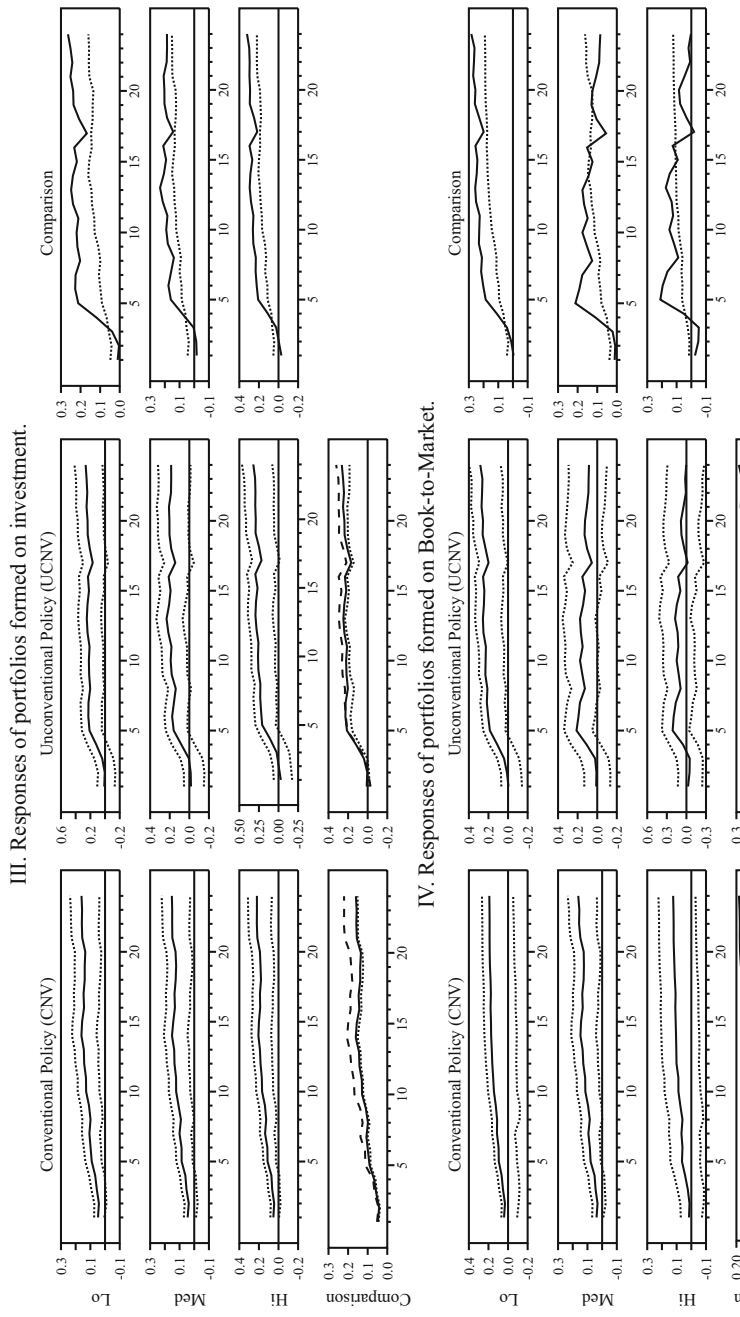


Fig. 6 continued

Out of two possible channels of transmissions, the empirical results lend support only to the interest rate channel. Based on this channel, we find that unconventional monetary policy has considerably higher effects on US stock returns than the conventional policy. We extended the analysis on the responses of portfolio returns in the US to two types of policy shocks and find supports for the main findings of the paper.

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